

Assignment 4.3 Bonus Question

Done by: Mohamed Walid

ID: 202202517

Code:

```
CREATE TABLE AIRPORT (  
    Airport_code VARCHAR(3) PRIMARY KEY,  
    Name VARCHAR(100),  
    City VARCHAR(50),  
    State VARCHAR(2)  
);  
  
INSERT INTO AIRPORT (Airport_code, Name, City, State) VALUES  
(  
    'JFK', 'John F. Kennedy International Airport', 'New York', 'NY'),  
(  
    'LAX', 'Los Angeles International Airport', 'Los Angeles', 'CA'),  
(  
    'ORD', 'O Hare International Airport', 'Chicago', 'IL'),  
(  
    'DFW', 'Dallas/Fort Worth International Airport', 'Dallas', 'TX'),  
(  
    'MIA', 'Miami International Airport', 'Miami', 'FL'),  
(  
    'SFO', 'San Francisco International Airport', 'San Francisco', 'CA'),  
(  
    'ATL', 'Hartsfield-Jackson Atlanta International Airport', 'Atlanta', 'GA');  
  
SELECT * FROM AIRPORT;
```

```
CREATE TABLE AIRPLANE_TYPE (  
    Airplane_type_name VARCHAR(50) PRIMARY KEY,  
    Max_seats INT,  
    Company VARCHAR(50)
```

);

```
INSERT INTO AIRPLANE_TYPE (Airplane_type_name, Max_seats, Company)
VALUES
```

```
('Boeing-737', 180, 'Boeing'),
```

```
('Airbus-A320', 150, 'Airbus'),
```

```
('Boeing-777', 350, 'Boeing'),
```

```
('Embraer-E190', 100, 'Embraer'),
```

```
('Airbus-A380', 500, 'Airbus');
```

```
SELECT * FROM AIRPLANE_TYPE;
```

```
CREATE TABLE CAN_LAND (
```

```
    Airplane_type_name VARCHAR(50),
```

```
    Airport_code VARCHAR(3),
```

```
    PRIMARY KEY (Airplane_type_name, Airport_code),
```

```
    FOREIGN KEY (Airplane_type_name) REFERENCES
AIRPLANE_TYPE(Airplane_type_name),
```

```
    FOREIGN KEY (Airport_code) REFERENCES AIRPORT(Airport_code)
```

```
);
```

```
INSERT INTO CAN_LAND (Airplane_type_name, Airport_code) VALUES
```

```
('Boeing-737', 'JFK'),
```

```
('Boeing-737', 'LAX'),
```

```
('Airbus-A320', 'ORD'),
```

```
('Boeing-777', 'DFW'),
```

```
('Embraer-E190', 'MIA'),
```

```
('Airbus-A380', 'SFO'),
```

```
('Airbus-A320', 'ATL');  
SELECT * FROM CAN_LAND;
```

```
CREATE TABLE AIRPLANE (  
    Airplane_id VARCHAR(10) PRIMARY KEY,  
    Total_number_of_seats INT,  
    Airplane_type VARCHAR(50),  
    FOREIGN KEY (Airplane_type) REFERENCES  
    AIRPLANE_TYPE(Airplane_type_name)  
);
```

```
INSERT INTO AIRPLANE (Airplane_id, Total_number_of_seats,  
    Airplane_type) VALUES  
('N12345', 180, 'Boeing-737'),  
('N67890', 150, 'Airbus-A320'),  
('N11223', 350, 'Boeing-777'),  
('N44556', 100, 'Embraer-E190'),  
('N77889', 500, 'Airbus-A380');  
SELECT * FROM AIRPLANE;
```

```
CREATE TABLE FLIGHT (  
    Flight_number VARCHAR(10) PRIMARY KEY,  
    Airline VARCHAR(50),  
    Weekdays VARCHAR(100)  
);  
INSERT INTO FLIGHT (Flight_number, Airline, Weekdays) VALUES  
('AA100', 'American Airlines', 'Monday,Wednesday,Friday'),
```

```
('UA200', 'United Airlines', 'Tuesday,Thursday,Saturday'),  
( 'DL300', 'Delta Air Lines',  
'Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday'),  
( 'SW400', 'Southwest Airlines', 'Monday,Friday,Sunday'),  
( 'JB500', 'JetBlue Airways', 'Wednesday,Saturday');  
SELECT * FROM FLIGHT;
```

```
CREATE TABLE FARE (  
    Flight_number VARCHAR(10),  
    Fare_code VARCHAR(10),  
    Amount DECIMAL(10,2),  
    Restrictions VARCHAR(100),  
    PRIMARY KEY (Flight_number, Fare_code),  
    FOREIGN KEY (Flight_number) REFERENCES FLIGHT(Flight_number)  
);
```

```
INSERT INTO FARE (Flight_number, Fare_code, Amount, Restrictions)  
VALUES
```

```
('AA100', 'ECONOMY', 299.99, 'No refund'),  
( 'AA100', 'BUSINESS', 599.99, '24hr cancellation'),  
( 'UA200', 'ECONOMY', 199.99, 'No changes'),  
( 'DL300', 'ECONOMY', 159.99, 'Basic economy'),  
( 'DL300', 'FIRST', 899.99, 'Fully flexible'),  
( 'SW400', 'ECONOMY', 129.99, 'No restrictions'),  
( 'JB500', 'ECONOMY', 179.99, '7-day advance');
```

```
SELECT * FROM FARE;
```

```
CREATE TABLE FLIGHT_LEG (  
    Flight_number VARCHAR(10),
```

```

    Leg_number INT,
    Departure_airport_code VARCHAR(3),
    Scheduled_departure_time TIME,
    Arrival_airport_code VARCHAR(3),
    Scheduled_arrival_time TIME,
    PRIMARY KEY (Flight_number, Leg_number),
    FOREIGN KEY (Flight_number) REFERENCES FLIGHT(Flight_number),
    FOREIGN KEY (Departure_airport_code) REFERENCES
AIRPORT(Airport_code),
    FOREIGN KEY (Arrival_airport_code) REFERENCES
AIRPORT(Airport_code)
);

```

```

INSERT INTO FLIGHT_LEG (Flight_number, Leg_number,
Departure_airport_code, Scheduled_departure_time, Arrival_airport_code,
Scheduled_arrival_time) VALUES
('AA100', 1, 'JFK', '08:00:00', 'LAX', '11:00:00'),
('UA200', 1, 'ORD', '09:30:00', 'DFW', '11:45:00'),
('DL300', 1, 'ATL', '07:15:00', 'MIA', '09:00:00'),
('SW400', 1, 'LAX', '06:45:00', 'SFO', '08:15:00'),
('JB500', 1, 'JFK', '14:20:00', 'MIA', '17:10:00');
SELECT * FROM FLIGHT_LEG;

```

```

CREATE TABLE LEG_INSTANCE (
    Flight_number VARCHAR(10),
    Leg_number INT,
    Date DATE,

```

```

Number_of_available_seats INT,
Airplane_id VARCHAR(10),
Departure_airport_code VARCHAR(3),
Departure_time TIME,
Arrival_airport_code VARCHAR(3),
Arrival_time TIME,
PRIMARY KEY (Flight_number, Leg_number, Date),
FOREIGN KEY (Flight_number, Leg_number) REFERENCES
FLIGHT_LEG(Flight_number, Leg_number),
FOREIGN KEY (Airplane_id) REFERENCES AIRPLANE(Airplane_id),
FOREIGN KEY (Departure_airport_code) REFERENCES
AIRPORT(Airport_code),
FOREIGN KEY (Arrival_airport_code) REFERENCES
AIRPORT(Airport_code)
);

INSERT INTO LEG_INSTANCE (Flight_number, Leg_number, Date,
Number_of_available_seats, Airplane_id, Departure_airport_code,
Departure_time, Arrival_airport_code, Arrival_time) VALUES
('AA100', 1, '2024-01-15', 150, 'N12345', 'JFK', '08:05:00', 'LAX', '11:10:00'),
('UA200', 1, '2024-01-16', 120, 'N67890', 'ORD', '09:35:00', 'DFW', '11:50:00'),
('DL300', 1, '2024-01-17', 300, 'N11223', 'ATL', '07:20:00', 'MIA', '09:05:00'),
('SW400', 1, '2024-01-18', 180, 'N44556', 'LAX', '06:50:00', 'SFO', '08:25:00'),
('JB500', 1, '2024-01-19', 95, 'N77889', 'JFK', '14:25:00', 'MIA', '17:15:00');

SELECT * FROM LEG_INSTANCE;

CREATE TABLE SEAT_RESERVATION (
    Flight_number VARCHAR(10),
    Leg_number INT,
    Date DATE,

```

```
Seat_number VARCHAR(5),  
Customer_name VARCHAR(100),  
Customer_phone VARCHAR(15),  
PRIMARY KEY (Flight_number, Leg_number, Date, Seat_number),  
FOREIGN KEY (Flight_number, Leg_number, Date) REFERENCES  
LEG_INSTANCE(Flight_number, Leg_number, Date)  
);
```

```
INSERT INTO SEAT_RESERVATION (Flight_number, Leg_number, Date,  
Seat_number, Customer_name, Customer_phone) VALUES  
( 'AA100', 1, '2024-01-15', '12A', 'John Smith', '555-0101'),  
( 'AA100', 1, '2024-01-15', '12B', 'Jane Smith', '555-0101'),  
( 'UA200', 1, '2024-01-16', '08C', 'Robert Johnson', '555-0102'),  
( 'DL300', 1, '2024-01-17', '15D', 'Maria Garcia', '555-0103'),  
( 'SW400', 1, '2024-01-18', '22F', 'David Brown', '555-0104'),  
( 'JB500', 1, '2024-01-19', '05A', 'Sarah Wilson', '555-0105');  
SELECT * FROM SEAT_RESERVATION;
```

Outputs:

1, CREATE TABLE AIRPORT (
2 Airport_code VARCHAR(3) PRIMARY KEY,
3 Name VARCHAR(100),
4 City VARCHAR(50),
5 State VARCHAR(2)
6);
7 INSERT INTO AIRPORT (Airport_code, Name, City, State) VALUES
8 ('JFK', 'John F. Kennedy International Airport', 'New York', 'NY'),
9 ('LAX', 'Los Angeles International Airport', 'Los Angeles', 'CA'),
10 ('ORD', 'O Hare International Airport', 'Chicago', 'IL'),
11 ('DFW', 'Dallas/Fort Worth International Airport', 'Dallas', 'TX'),
12 ('MIA', 'Miami International Airport', 'Miami', 'FL'),
13 ('SFO', 'San Francisco International Airport', 'San Francisco', 'CA'),
14 ('ATL', 'Hartsfield-Jackson Atlanta International Airport', 'Atlanta', 'GA');
15 SELECT * FROM AIRPORT;
16
17 CREATE TABLE AIRPLANE_TYPE (
18 Airplane_type_name VARCHAR(50) PRIMARY KEY,
19 Max_seats INT,
20 Company VARCHAR(50)
21);
22
23 INSERT INTO AIRPLANE_TYPE (Airplane_type_name, Max_seats, Company) VALUES
24 ('Boeing-737', 180, 'Boeing'),
25 ('Airbus-A320', 150, 'Airbus'),
26 ('Boeing-777', 350, 'Boeing'),
27 ('Embraer-E190', 100, 'Embraer'),
28 ('Airbus-A380', 500, 'Airbus');
29 SELECT * FROM AIRPLANE_TYPE;
30
31 CREATE TABLE CAN_LAND (
32 Airplane_type_name VARCHAR(50),
33 Airport_code VARCHAR(3),

Records: 7 Duplicates: 0 Warnings: 0

Airport_code	Name	City	State
ATL	Hartsfield-Jackson Atlanta International Airport	Atlanta	GA
DFW	Dallas/Fort Worth International Airport	Dallas	TX
JFK	John F. Kennedy International Airport	New York	NY
LAX	Los Angeles International Airport	Los Angeles	CA
MIA	Miami International Airport	Miami	FL
ORD	O Hare International Airport	Chicago	IL
SFO	San Francisco International Airport	San Francisco	CA

Records: 5 Duplicates: 0 Warnings: 0

Airplane_type_name	Max_seats	Company
Airbus-A320	150	Airbus
Airbus-A380	500	Airbus
Boeing-737	180	Boeing
Boeing-777	350	Boeing
Embraer-E190	100	Embraer

Records: 7 Duplicates: 0 Warnings: 0

Airplane_type_name	Airport_code
Airbus-A320	ATL
Boeing-777	DFW
Boeing-737	JFK
Boeing-737	LAX
Embraer-E190	MIA
Airbus-A320	ORD
Airbus-A380	SFO

Activate Windows
Go to Settings to activate Windows.

31 CREATE TABLE CAN_LAND (
32 Airplane_type_name VARCHAR(50),
33 Airport_code VARCHAR(3),
34 PRIMARY KEY (Airplane_type_name, Airport_code),
35 FOREIGN KEY (Airplane_type_name) REFERENCES AIRPLANE_TYPE (Airplane_type_name),
36 FOREIGN KEY (Airport_code) REFERENCES AIRPORT (Airport_code)
37);
38
39 INSERT INTO CAN_LAND (Airplane_type_name, Airport_code) VALUES
40 ('Boeing-737', 'JFK'),
41 ('Boeing-737', 'LAX'),
42 ('Airbus-A320', 'ORD'),
43 ('Boeing-777', 'DFW'),
44 ('Embraer-E190', 'MIA'),
45 ('Airbus-A380', 'SFO'),
46 ('Airbus-A320', 'ATL');
47 SELECT * FROM CAN_LAND;
48
49 CREATE TABLE AIRPLANE (
50 Airplane_id VARCHAR(10) PRIMARY KEY,
51 Total_number_of_seats INT,
52 Airplane_type VARCHAR(50),
53 FOREIGN KEY (Airplane_type) REFERENCES AIRPLANE_TYPE (Airplane_type_name)
54);
55
56 INSERT INTO AIRPLANE (Airplane_id, Total_number_of_seats, Airplane_type) VALUES
57 ('N12345', 180, 'Boeing-737'),
58 ('N67890', 150, 'Airbus-A320'),
59 ('N11223', 350, 'Boeing-777'),
60 ('N44556', 100, 'Embraer-E190'),
61 ('N77889', 500, 'Airbus-A380');
62 SELECT * FROM AIRPLANE;
63
64 CREATE TABLE FLIGHT (
65 Flight_number VARCHAR(10) PRIMARY KEY,

Records: 5 Duplicates: 0 Warnings: 0

Airplane_id	Total_number_of_seats	Airplane_type
N11223	350	Boeing-777
N12345	180	Boeing-737
N44556	100	Embraer-E190
N67890	150	Airbus-A320
N77889	500	Airbus-A380

Records: 5 Duplicates: 0 Warnings: 0

Flight_number	Airline	Weekdays
AA100	American Airlines	Monday,Wednesday,Friday
DL300	Delta Air Lines	Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday
JB500	JetBlue Airways	Wednesday,Saturday
SW400	Southwest Airlines	Monday,Friday,Sunday
UA200	United Airlines	Tuesday,Thursday,Saturday

Records: 7 Duplicates: 0 Warnings: 0

Flight_number	Fare_code	Amount	Restrictions
AA100	BUSINESS	599.99	24hr cancellation
AA100	ECONOMY	299.99	No refund
DL300	ECONOMY	159.99	Basic economy
DL300	FIRST	899.99	Fully flexible
JB500	ECONOMY	179.99	7-day advance
SW400	ECONOMY	129.99	No restrictions
UA200	ECONOMY	199.99	No changes

Activate Windows
Go to Settings to activate Windows.

49 CREATE TABLE AIRPLANE (
50 Airplane_id VARCHAR(10) PRIMARY KEY,
51 Total_number_of_seats INT,
52 Airplane_type VARCHAR(50),
53 FOREIGN KEY (Airplane_type) REFERENCES AIRPLANE_TYPE (Airplane_type_name)
54);
55
56 INSERT INTO AIRPLANE (Airplane_id, Total_number_of_seats, Airplane_type) VALUES
57 ('N12345', 180, 'Boeing-737'),
58 ('N67890', 150, 'Airbus-A320'),
59 ('N11223', 350, 'Boeing-777'),
60 ('N44556', 100, 'Embraer-E190'),
61 ('N77889', 500, 'Airbus-A380');
62 SELECT * FROM AIRPLANE;
63
64 CREATE TABLE FLIGHT (
65 Flight_number VARCHAR(10) PRIMARY KEY,
66 Airline VARCHAR(50),
67 Weekdays VARCHAR(100)
68);
69 INSERT INTO FLIGHT (Flight_number, Airline, Weekdays) VALUES
70 ('AA100', 'American Airlines', 'Monday,Wednesday,Friday'),
71 ('UA200', 'United Airlines', 'Tuesday,Thursday,Saturday'),
72 ('DL300', 'Delta Air Lines', 'Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday'),
73 ('SW400', 'Southwest Airlines', 'Monday,Friday,Sunday'),
74 ('JB500', 'JetBlue Airways', 'Wednesday,Saturday');
75 SELECT * FROM FLIGHT;
76
77 CREATE TABLE FARE (
78 Flight_number VARCHAR(10),
79 Fare_code VARCHAR(10),
80 Amount DECIMAL(10,2),
81 Restrictions VARCHAR(100),
82 PRIMARY KEY (Flight_number, Fare_code),
83 FOREIGN KEY (Flight_number) REFERENCES FLIGHT (Flight_number)
84);

Records: 7 Duplicates: 0 Warnings: 0

Flight_number	Leg_number	Departure_airport_code	Scheduled_departure_time	Arrival_airport_code	Scheduled_arrival_time
AA100	1	JFK	08:00:00	LAX	11:00:00
DL300	1	ATL	07:15:00	MIA	09:00:00
JB500	1	JFK	14:20:00	MIA	17:10:00
SW400	1	LAX	06:45:00	SFO	08:15:00
UA200	1	ORD	09:30:00	DFW	11:45:00

Records: 5 Duplicates: 0 Warnings: 0

Flight_number	Leg_number	Date	Number_of_available_seats	Airplane_id	Departure_airport_code	Departure_time	Arrival
AA100	1	2024-01-15	150	N12345	JFK	08:05:00	LAX
DL300	1	2024-01-17	300	N11223	ATL	07:20:00	MIA
JB500	1	2024-01-19	95	N77889	JFK	14:25:00	MIA
SW400	1	2024-01-18	180	N44556	LAX	06:50:00	SFO
UA200	1	2024-01-16	120	N67890	ORD	09:35:00	DFW

Activate Windows
Go to Settings to activate Windows.

65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
101

```
65 -- Foreign key constraint (FK) Flight-Fare
66 Airline VARCHAR(50),
67 Weekdays VARCHAR(100)
68 );
69 INSERT INTO FLIGHT (Flight_number, Airline, Weekdays) VALUES
70 ('AA100', 'American Airlines', 'Monday,Wednesday,Friday'),
71 ('UA200', 'United Airlines', 'Tuesday,Thursday,Saturday'),
72 ('DL300', 'Delta Air Lines', 'Monday,Tuesday,Wednesday,Thursday,Friday,Saturday,Sunday'),
73 ('SW400', 'Southwest Airlines', 'Monday,Friday,Sunday'),
74 ('JB500', 'JetBlue Airways', 'Wednesday,Saturday');
75 SELECT * FROM FLIGHT;
76
77 -- Create table FARE (
78 Flight_number VARCHAR(10),
79 Fare_code VARCHAR(10),
80 Amount DECIMAL(10,2),
81 Restrictions VARCHAR(100),
82 PRIMARY KEY (Flight_number, Fare_code),
83 FOREIGN KEY (Flight_number) REFERENCES FLIGHT(Flight_number)
84 );
85 INSERT INTO FARE (Flight_number, Fare_code, Amount, Restrictions) VALUES
86 ('AA100', 'ECONOMY', 299.99, 'No refund'),
87 ('AA100', 'BUSINESS', 599.99, '24hr cancellation'),
88 ('UA200', 'ECONOMY', 199.99, 'No changes'),
89 ('DL300', 'ECONOMY', 159.99, 'Basic economy'),
90 ('DL300', 'FIRST', 899.99, 'Fully flexible'),
91 ('SW400', 'ECONOMY', 129.99, 'No restrictions'),
92 ('JB500', 'ECONOMY', 179.99, '7-day advance');
93 SELECT * FROM FARE;
94 -- Create table FLIGHT_LEG (
95 Flight_number VARCHAR(10),
96 Leg_number INT,
97 Departure_airport_code VARCHAR(3),
98 Scheduled_departure_time TIME,
99 Arrival_airport_code VARCHAR(3),
100 Scheduled_arrival_time TIME,
101 PRIMARY KEY (Flight_number, Leg_number),
```

Records: 5 Duplicates: 0 Warnings: 0

Flight_number	Leg_number	Departure_airport_code	Scheduled_departure_time	Arrival_airport_code	Scheduled_arrival_time
AA100	1	JFK	08:00:00	LAX	11:00:00
DL300	1	ATL	07:15:00	MIA	09:00:00
JB500	1	JFK	14:20:00	MIA	17:10:00
SW400	1	LAX	06:45:00	SFO	08:15:00
UA200	1	ORD	09:30:00	DFW	11:45:00

Records: 5 Duplicates: 0 Warnings: 0

Flight_number	Leg_number	Date	Number_of_available_seats	Airplane_id	Departure_airport_code	Departure_time	Arrival_airport_code	Arrival_time
AA100	1	2024-01-15	150	N12345	JFK	08:05:00	LAX	11:05:00
DL300	1	2024-01-17	300	N11223	ATL	07:20:00	MIA	09:20:00
JB500	1	2024-01-19	95	N77889	JFK	14:25:00	MIA	17:15:00
SW400	1	2024-01-18	180	N44556	LAX	06:50:00	SFO	08:20:00
UA200	1	2024-01-16	120	N67890	ORD	09:35:00	DFW	11:50:00

Records: 6 Duplicates: 0 Warnings: 0

Flight_number	Leg_number	Date	Seat_number	Customer_name	Customer_phone
AA100	1	2024-01-15	12A	John Smith	555-0101
AA100	1	2024-01-15	12B	Jane Smith	555-0101
DL300	1	2024-01-17	15D	Maria Garcia	555-0103
JB500	1	2024-01-19	05A	Sarah Wilson	555-0105
SW400	1	2024-01-18	22F	David Brown	555-0104
UA200	1	2024-01-16	08C	Robert Johnson	555-0102

Activate Windows
Go to Settings to activate Windows.